Regular Grammars

Computation and Discrete Structures III

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Contenido

1. Regular Grammars

Languages, Machine Types and Grammars

Type	Languages	Machine Type	Grammar Rules
0	Recursively enumerable	Turing machine	Unrestricted
1	Context-sensitive	Linear bounded automaton	$\alpha \to \beta, \quad \lvert\alpha\rvert \leq \lvert\beta\rvert$
2	Context-free	Pushdown automaton	$A \to \gamma$
3	Regular	Finite automaton	$A \to aB$ $A \to a$

Regular Grammars

$$S \rightarrow aA \mid bB$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow bB \mid b$$

- S, A and B are non-terminals, indicating they must be replaced according to the productions
- a and b are terminals belonging to an alphabet Σ

Regular Grammars

$$S \rightarrow aA \mid bB$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow bB \mid b$$

Grammars generate strings

Regular Grammars

$$S \rightarrow aA \mid bB$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow bB \mid b$$

Grammars generate strings

$S \rightarrow aA \rightarrow aaA \rightarrow aaa$

The string aaa is generated by the grammar

Regular Grammars

$$S \rightarrow aA \mid bB$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow bB \mid b$$

$$S \rightarrow aA \rightarrow aa$$
$$S \rightarrow aA \rightarrow aaA \rightarrow aaaA \rightarrow aaaa$$
$$S \rightarrow bB \rightarrow bb$$

Regular Grammars

$$S \rightarrow aA \mid bB$$
$$A \rightarrow aA \mid a$$
$$B \rightarrow bB \mid b$$

Can the string *ab* be generated by the grammar?

Regular Grammars

$$S \rightarrow aAb \mid bBa$$
$$A \rightarrow aAb \mid \epsilon$$
$$B \rightarrow bBa \mid \epsilon$$

Indicate which of the following strings can be generated by the grammar:

- ab
- $aabb$
- $bbaa$
- abb
- $bbba$

Regular Grammars

$$S \rightarrow abS \mid \epsilon$$

Indicate which of the following strings can be generated by the grammar:

- ϵ
- $abab$
- $aaab$
- abb

Regular Grammars

$$S \rightarrow aE$$
$$E \rightarrow A \mid B$$
$$A \rightarrow aA \mid b$$
$$B \rightarrow aB \mid b$$

The string $aaab$ can be generated as:
$S \rightarrow aE \rightarrow aA \rightarrow aaA \rightarrow aaaA \rightarrow aaab$
The notation $S \xrightarrow{*} w$ indicates that string w **can be generated** from S in 0 or more steps

Regular Grammars

Consider the regular language $a(a^* \cup b^*)b$. One way to express the strings accepted by the language is through the productions

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Consider the regular language $a(a^* \cup b^*)b$. One way to express the strings accepted by the language is through the productions

$$S \to aE$$

$$E \to A \mid B$$

$$A \to aA \mid b$$

$$B \to aB \mid b$$

Regular Grammars

A **regular grammar** is defined as a set of 4 elements, $G = (\Sigma, N, S, P)$ where:

- Σ is the alphabet
- N are the non-terminals
- S is the start symbol
- P is the collection of substitution rules or productions

$$S \rightarrow aE$$
$$E \rightarrow A \mid B$$
$$A \rightarrow aA \mid b$$
$$B \rightarrow aB \mid b$$

Regular Grammars

A **regular grammar** is defined as a set of 4 elements, $G = (\Sigma, N, S, P)$ where:

- Σ is the alphabet
- N are the non-terminals
- S is the start symbol
- P is the collection of substitution rules or productions of the form $A \rightarrow w$, where $A \in N$ and $w \in (\Sigma \cup N)^*$ that satisfies:
  1. w contains at most one non-terminal
  2. If w contains a non-terminal, it is the rightmost symbol in w

Regular Grammars

The following grammars are not regular:

$$S \to AB$$
$$A \to aA \mid a$$
$$B \to bB \mid b$$

$$S \to aAb$$
$$A \to cA \mid c$$

$$aSb \to aA$$
$$bAa \to b \mid c$$

Regular Grammars

Consider the following regular grammar:

- $\Sigma = \{a, b\}$
- $N = \{S, A\}$
- S is the start symbol
- P :

$$S \rightarrow bA$$
$$A \rightarrow aaA \mid b$$

Regular Grammars

Consider the following regular grammar:

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- $N = \{S, A\}$
- S is the start symbol
- P :

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$$A \rightarrow aaA \mid b$$

$bb, baab, baaaab, baaaaaab, \ldots$

Regular Grammars

Consider the following regular grammar:

- $\Sigma = \{a, b\}$
- $N = \{S, A\}$
- S is the start symbol
- P :

$$S \rightarrow bA$$
$$A \rightarrow aaA \mid b$$

The language accepted by the grammar, $L(G)$, contains strings of the form $b(aa)^*b$

Regular Grammars

Indicate the regular expression associated with the following grammar:

- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow aS \mid b$$

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Indicate the regular expression associated with the following grammar:

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- S is the start symbol
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$$B \rightarrow bB \mid \epsilon$$

Regular Grammars

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The language accepted by the grammar, $L(G)$, contains strings of the form a^*b^*

Regular Grammars

Indicate the regular expression associated with the following grammar:

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- $N = \{S, A\}$
- S is the start symbol
- P :

$$S \rightarrow abS \mid A$$
$$A \rightarrow a \mid b$$

Regular Grammars

Indicate the regular expression associated with the following grammar:

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The language accepted by the grammar, $L(G)$, contains strings of the form $(ab)^*(a \cup b)$

Regular Grammars

Design a regular grammar that recognizes $(ab)^+$

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- $\Sigma = \{a, b\}$
- $N = \{S\}$
- S is the start symbol
- P :

$$S \rightarrow abS \mid ab$$

Regular Grammars

Design a regular grammar that recognizes the language given by $(a \cup b)a^*(a \cup b)$

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- $N = \{S, A\}$
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Design a regular grammar that recognizes the language given by $(a \cup b)^* a (a \cup b)^*$

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- $\Sigma = \{a, b\}$
- $N = \{S, A\}$
- S is the start symbol
- P :

$$S \rightarrow aS \mid bS \mid aA$$
$$A \rightarrow aA \mid bA \mid \epsilon$$

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- S is the start symbol
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$$S \rightarrow abS \mid abA$$
$$A \rightarrow aA \mid bA \mid \epsilon$$

Regular Grammars

Design a regular grammar that recognizes the language given by $a^*b^*c^*$

Regular Grammars

Design a regular grammar that recognizes the language given by $a^*b^*c^*$

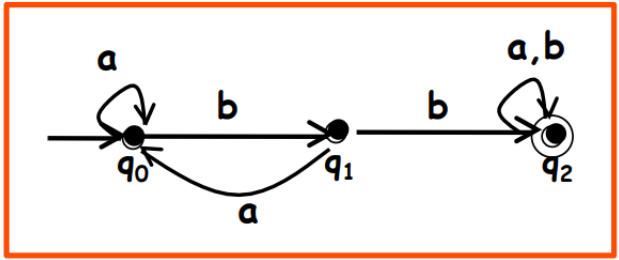
- $\Sigma = \{a, b, c\}$
- $N = \{S, B, C\}$
- S is the start symbol
- P :

$$
\begin{aligned}
S &\rightarrow aS \mid bB \mid cC \mid \epsilon \\
B &\rightarrow bB \mid cC \mid \epsilon \\
C &\rightarrow cC \mid \epsilon
\end{aligned}
$$

Regular Grammars

Theorem

Given an automaton M, there exists a grammar G such that $L(M) = L(G)$



$$S \rightarrow aS \mid bA$$

$$A \rightarrow aS \mid bB$$

$$B \rightarrow aB \mid bB \mid \varepsilon$$

Regular Grammars

Theorem

Given an automaton M, there exists a grammar G such that $L(M) = L(G)$

The productions are obtained by taking the automaton states as non-terminals and alphabet symbols as terminals

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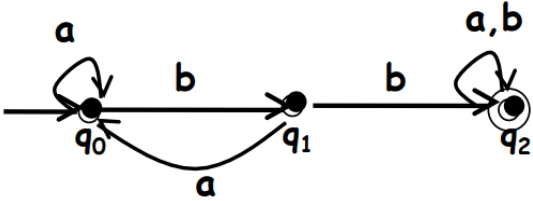

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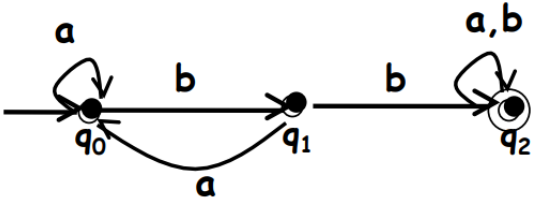
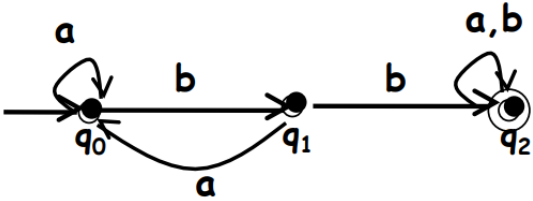


Regular Grammars



The automaton *M* induces the regular grammar:

$q_0 \rightarrow aq_0 | bq_1$

$q_1 \rightarrow aq_0 | bq_2$

$q_2 \rightarrow aq_2 | bq_2 | \varepsilon$

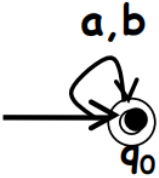
$S \rightarrow aS | bA$

$A \rightarrow aS | bB$

$B \rightarrow aB | bB | \varepsilon$

Regular Grammars

Automaton that recognizes $(a \cup b)^*$



Regular Grammars

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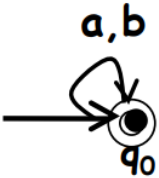
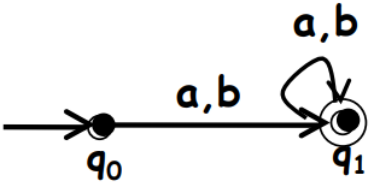


The automaton M induces the regular grammar:

$$q_0 \rightarrow aq_0 | bq_0 | \varepsilon \quad \Rightarrow \quad S \rightarrow aS | bS | \varepsilon$$

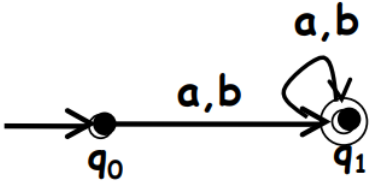
Regular Grammars

Automaton that recognizes $(a \cup b)^+$



Regular Grammars

Automaton that recognizes $(a \cup b)^+$



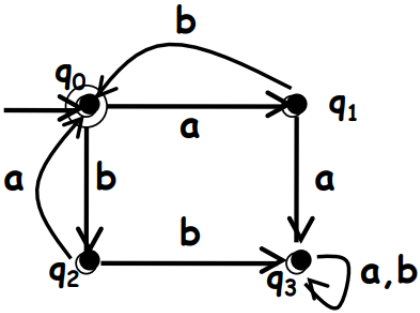
The automaton M induces the regular grammar:

$$q_0 \rightarrow aq_1 | bq_1$$
$$q_1 \rightarrow aq_1 | bq_1 | \varepsilon$$

$$S \rightarrow aA | bA$$
$$A \rightarrow aA | bA | \varepsilon$$

Regular Grammars

Show the regular grammar for the following automaton that recognizes $(ab \cup ba)^*$



Regular Grammars

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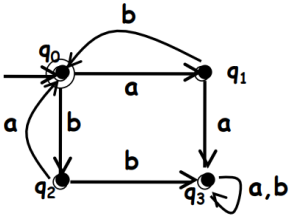


The automaton M induces the regular grammar:

$q_0 \rightarrow aq_1 | bq_2 | \varepsilon$

$q_1 \rightarrow bq_0 | aq_3$

$q_2 \rightarrow aq_0 | bq_3$

$q_3 \rightarrow aq_3 | bq_3$

$S \rightarrow aA | bB | \varepsilon$

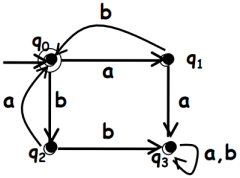
$A \rightarrow bS | aC$

$B \rightarrow aS | bC$

$C \rightarrow aC | bC$

Regular Grammars

Show the regular grammar for the following automaton that recognizes $(ab \cup ba)^*$



The automaton M induces the regular grammar:

$q_0 \rightarrow aq_1 | bq_2 | \varepsilon$

$q_1 \rightarrow bq_0 | aq_3$

$q_2 \rightarrow aq_0 | bq_3$

$q_3 \rightarrow aq_3 | bq_3$

$S \rightarrow aA | bB | \varepsilon$

$A \rightarrow bS | aC$

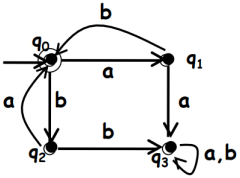
$B \rightarrow aS | bC$

$C \rightarrow aC | bC$

Evaluate the string *aab*

Regular Grammars

Show the regular grammar for the following automaton that recognizes $(ab \cup ba)^*$



The automaton M induces the regular grammar:

$q_0 \rightarrow aq_1 | bq_2 | \varepsilon$

$q_1 \rightarrow bq_0 | aq_3$

$q_2 \rightarrow aq_0 | bq_3$

$q_3 \rightarrow aq_3 | bq_3$

$S \rightarrow aA | bB | \varepsilon$

$A \rightarrow bS | aC$

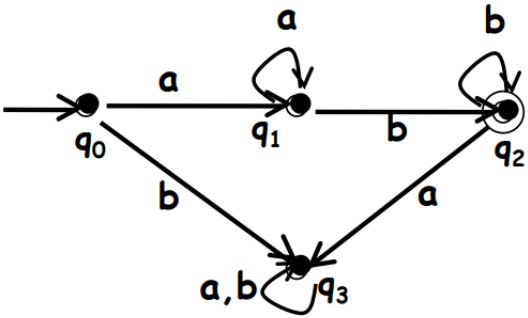
$B \rightarrow aS | bC$

$C \rightarrow aC | bC$

The string *aab* is not generated by the grammar

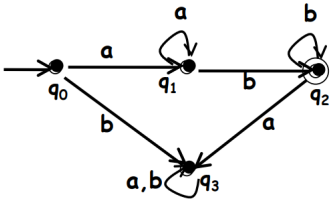
Regular Grammars

Show the regular grammar for the following automaton that recognizes a^+b^+



Regular Grammars

Show the regular grammar for the following automaton that recognizes a^+b^+



The automaton M induces the regular grammar:

$$
\begin{array}{l}
q_0 \rightarrow aq_1 | bq_3 \\
q_1 \rightarrow aq_1 | bq_2 \\
q_2 \rightarrow bq_2 | \varepsilon | aq_3 \\
q_3 \rightarrow aq_3 | bq_3
\end{array}
\quad \Rightarrow \quad
\begin{array}{l}
S \rightarrow aA | bC \\
A \rightarrow aA | bB \\
B \rightarrow bB | \varepsilon | aC \\
C \rightarrow aC | bC
\end{array}
$$

Regular Grammars

Theorem

Given a regular grammar G, there exists an automaton M such that $L(M) = L(G)$

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The automaton construction uses the productions, considering non-terminals as states and terminals as transitions

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Given a regular grammar G, there exists an automaton M such that $L(M) = L(G)$

The automaton construction uses the productions, considering non-terminals as states and terminals as transitions

$$S \rightarrow aS \mid bA$$
$$A \rightarrow aB \mid bB \mid \epsilon$$
$$B \rightarrow aB \mid bB$$

Regular Grammars

Design the automaton for the following grammar:

$$S \rightarrow aS \mid bA$$

$$A \rightarrow aB \mid bB \mid \epsilon$$

$$B \rightarrow aB \mid bB$$

Regular Grammars

Design the automaton for the following grammar:

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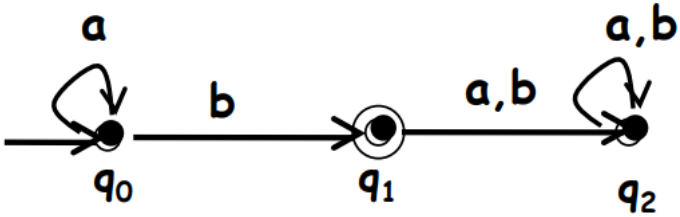
$A \rightarrow aB \mid bB \mid \epsilon$

$B \rightarrow aB \mid bB$



Regular Grammars

Design the automaton for the following grammar:

$$S \rightarrow aB \mid bA \mid \epsilon$$
$$A \rightarrow abaS$$
$$B \rightarrow babS$$

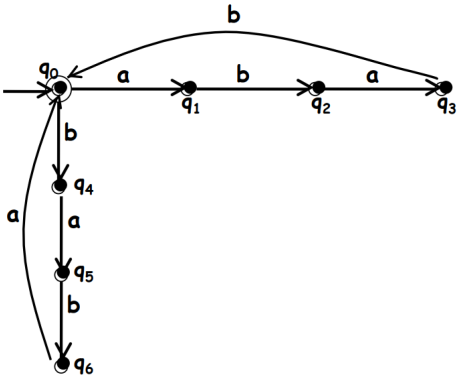
Regular Grammars

Design the automaton for the following grammar:

$S \rightarrow aB \mid bA \mid \epsilon$

$A \rightarrow abaS$

$B \rightarrow babS$



Regular Grammars

Design the automaton for the following grammar:

$$S \rightarrow aA \mid \epsilon$$
$$A \rightarrow abA \mid baB \mid \epsilon$$
$$B \rightarrow aB \mid bA$$

References

Kozen, D. C. (2007)
Automata and computability
Springer Science & Business Media.